

Computer Vision

Lecture 3

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Misc about contour

- `cv2.boundingRect(contour)` #Get the 4 points of the bounding rectangle
- `cv2.minAreaRect(contour)` #Get the 4 points of the bounding rectangle with the minimum area
- `cv2.convexHull(contour)` #Detect the convex contour
- https://docs.opencv.org/3.4.6/d3/d05/tutorial_py_table_of_contents_contours.html

Bitwise operation

- `cv2.bitwise_not(binary_pic)`
- `cv2.bitwise_and(img,img2,mask)`
- `cv2.bitwise_or(img,img2, mask)`
- `cv2.bitwise_xor(img, img2, mask)`

Pillow

- 安裝 pillow
- >> conda install -c anaconda pillow
- >> pip install Pillow
- <https://pillow.readthedocs.io>
- Support PNG , JPEG , PPM , GIF , TIFF , BMP

From PIL import Image

- Image.open(pathway)
- Image.save(pathway)
- Image.resize((width,height))
- Image.thumbnail ((x-axial length,y-axial length))#保持原本比例
- Image.size #return (width,hegit); Opencv return (height, width, channels)
- Image.crop((x1,y1,x2,y2))
- Image.copy()
- Image.paste(pic, (x,y))
- Image.rotate(degree, expend = False)
- Image.convert(flag)
 - Flag = '1', 'L', 'P', 'RGB', 'RGBA', 'CMYK', 'YCbCr', 'I', 'F'

From PIL import Image-2

- `Image.transpose(PIL.Image.FLIP_LEFT_RIGHT)`
 - `PIL.Image.FLIP_LEFT_RIGHT`
 - `PIL.Image.FLIP_TOP_BOTTOM`
 - `PIL.Image.ROTATE_90`
 - `PIL.Image.ROTATE_180`
 - `PIL.Image.ROTATE_270`
 - `PIL.Image.TRANSPOSE`

ImageDraw, ImageFont

- `img_draw = ImageDraw(canvas)`
- `img_draw.rectangle((x1,y1,x2,y2), outline=color, fill=color)`
- `fnt = ImageFont.truetype(fontFilePath, size)`
- `img_draw.text((x,y), text, font=fnt, fill=color)`
- `img_draw.line([(x1,y1), (x2,y2), ...], fill=color, width)`
- `img_draw.ellipse((270-r,200-r, 270+r, 200+r), fill=color)`
 #(270,200)為圓心, r為半徑
- `img_draw.polygon([(x1, y1), (x2, y2), (x3, y3), (x4, y4)], fill=color, outline=color)`

Facial Keypoints Recognition

Predicting the keypoint locations on facial images can be used as a building block in several applications/programs:

- Tracking facial positions in images and videos:
 - Analyzing facial expressions
 - Detecting signs of facial deformities for medical diagnosis
 - Biometric identification
 - Face recognition
- <https://www.kaggle.com/c/facial-keypoints-detection/data>

Kaggle facial keypoint dataset

- 7,049個 96x96 灰度圖像 組成
- 15 xy position pair per picture
- left_eye_center, right_eye_outer_corner, mouth_center_bottom_lip etc..
- Some key points have approximately 2,000 labels, while other key points have over 7,000 labels available for training.
 - For example,
 - left_eye_center_x has 7,034 entries,
 - left_eye_inner_corner_x has only 2,266 entries.